

Evaluating the Parabolic SAR Strategy on Bitcoin

Author: Bc. David Mesíček

Affiliation: Independent Researcher, Quantitative Trading Studies

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Abstract

This study evaluates the performance of the Parabolic Stop and Reverse (SAR) indicator when applied to Bitcoin (BTC/USDT) across two timeframes (1H and 1D) from 2015 to 2025. Three trading modes—BUYONLY, SELLONLY, and BOTH (long–short reversal)—are compared against the Bitcoin Buy & Hold benchmark.

Results show that Parabolic SAR, despite correctly identifying trend inflection points, performs poorly as a standalone system. BUYONLY produces moderate returns yet fails to outperform Buy & Hold. SELLONLY and BOTH modes exhibit catastrophic decay due to Bitcoin's strong structural upward drift and SAR's high sensitivity to whipsaws.

Seasonality-based calibration (restricting trading to profitable hours/days) improves drawdowns and volatility but cannot achieve overall profitability. Parabolic SAR requires additional filters—trend confirmation, volatility regimes, or multi-timeframe context—to function effectively in cryptocurrency markets.

1. Introduction

Bitcoin is one of the most volatile and structurally unique assets in global financial markets. Following the introduction of CME Bitcoin futures in 2018, BTC entered a more mature phase characterized by institutional market structure, deep liquidity, and repeatable volatility patterns. This makes it a suitable testing ground for systematic trading strategies and trend-following indicators.

Welles Wilder's Parabolic SAR is a classical indicator designed to track trend reversals by accelerating a trailing stop in the direction of price movement. While SAR is visually intuitive and effective at highlighting trend phases, its performance in the context of modern Bitcoin market microstructure is uncertain.

This study rigorously evaluates whether Parabolic SAR can serve as a profitable standalone system across intraday and daily horizons. We examine all structural modes (long-only, short-only, and reversal) and test seasonality-driven calibration windows based on the most profitable hours and days.

2. Goal and Motivation

The primary goal is to assess whether the Parabolic SAR can generate consistent, risk-adjusted returns on Bitcoin.

Specifically, the study aims to:

- Compare BUYONLY, SELLONLY, and BOTH modes to identify structural strengths and weaknesses.
- Quantify SAR performance on 1H vs 1D timeframes.
- Evaluate PSAR's sensitivity to intraday/weekday seasonality.
- Test whether calibration to statistically profitable windows improves performance.

Bitcoin's long-term upward drift suggests that long-only approaches may have structural advantages. However, this study examines all modes to develop a complete understanding of SAR behaviour.

3. Data

3.1 Dataset Description

- Asset: **BTC/USDT**
- Exchanges: Binance, Gemini
- Period: **2015 – October 2025**
- Timeframes: **1H and 1D**

3.2 Benchmark Performance (Buy & Hold)

Bitcoin Buy & Hold exhibits:

- extremely high long-term returns,
- catastrophic drawdowns,
- volatility consistent with high-risk assets.

[INSERT FIGURE: BTC Buy & Hold Equity Curve – benchmark graph]

Buy & Hold therefore serves as a high-return but high-risk baseline.

4. Methodology

4.1 Parabolic SAR Definition

$$SAR_{t+1} = SAR_t + \alpha (EP_t - SAR_t) \\ SAR_{t+1} = SAR_t + \alpha (EP_t - SAR_t)$$

- Step = **0.02**
- Max Step = **0.2**

4.2 Signal Logic

$$Signal = \begin{cases} 1, & \text{Close} > PSAR \\ -1, & \text{Close} < PSAR \end{cases} \\ Signal = \begin{cases} 1, & \text{Close} > PSAR \\ -1, & \text{Close} < PSAR \end{cases}$$

4.3 Execution Modes

1. **BUYONLY**: long when bullish, flat otherwise
2. **SELLONLY**: short when bearish, flat otherwise
3. **BOTH**: full reversal long/short

4.4 Backtesting Assumptions

- no fees
- no slippage
- full capital allocation
- equity normalized to 1
- trades at bar close

5. Results

5.1 1-Hour Timeframe

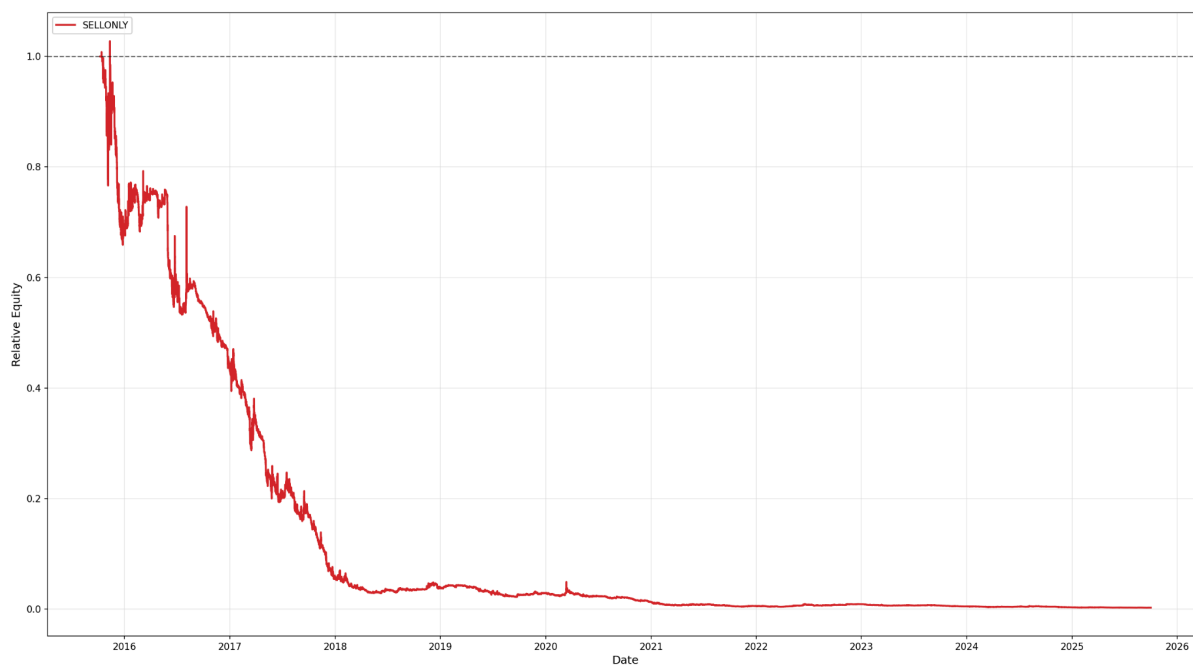
BUYONLY (1H)

Annualized Return: **35.95 %**
Maximum Drawdown: **-81.83 %**
Annualized Volatility: **49.69 %**
Sharpe Ratio: **0.72**
Calmar Ratio: **0.44**
Number of Trades: **6,370**



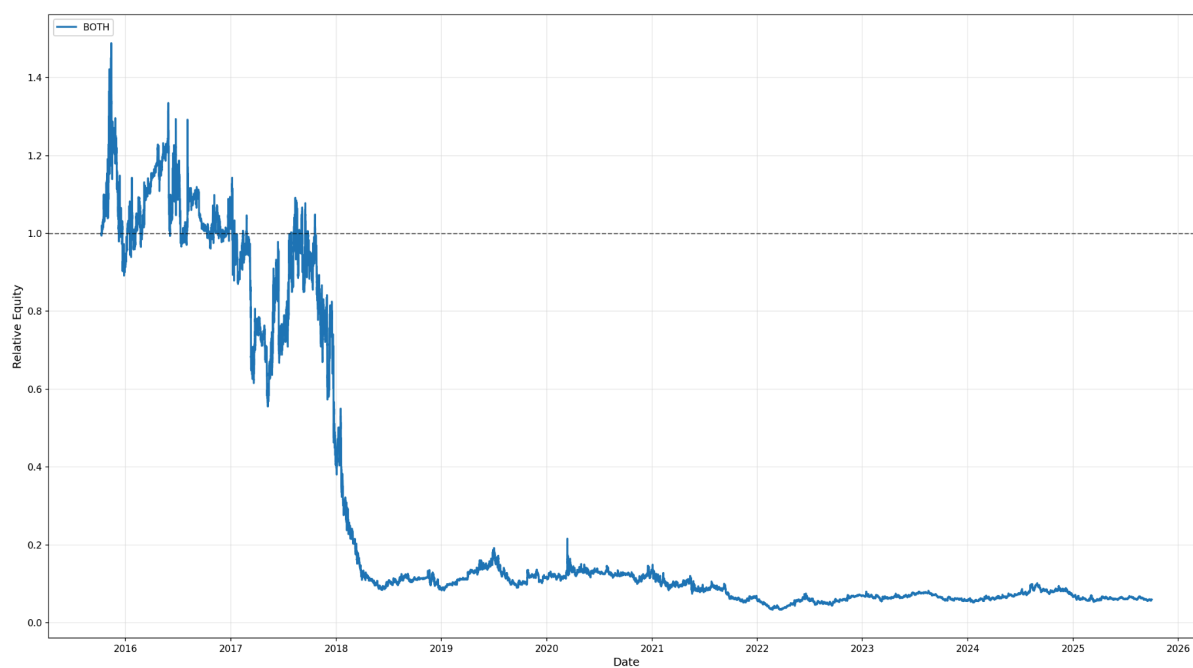
SELLONLY (1H)

Annualized Return: **-44.55 %**
Maximum Drawdown: **-99.74 %**
Annualized Volatility: **53.27 %**
Sharpe Ratio: **-0.84**
Calmar Ratio: **-0.45**
Number of Trades: **6,382**



BOTH (1H)

Annualized Return: **-24.62 %**
Maximum Drawdown: **-97.75 %**
Annualized Volatility: **72.85 %**
Sharpe Ratio: **-0.34**
Calmar Ratio: **-0.25**
Number of Trades: **12,752**



5.2 1-Day Timeframe

BUYONLY (1D)

Annualized Return: **46.94 %**
Maximum Drawdown: **-66.73 %**
Annualized Volatility: **46.96 %**
Sharpe Ratio: **1.00**
Calmar Ratio: **0.70**
Number of Trades: **292**



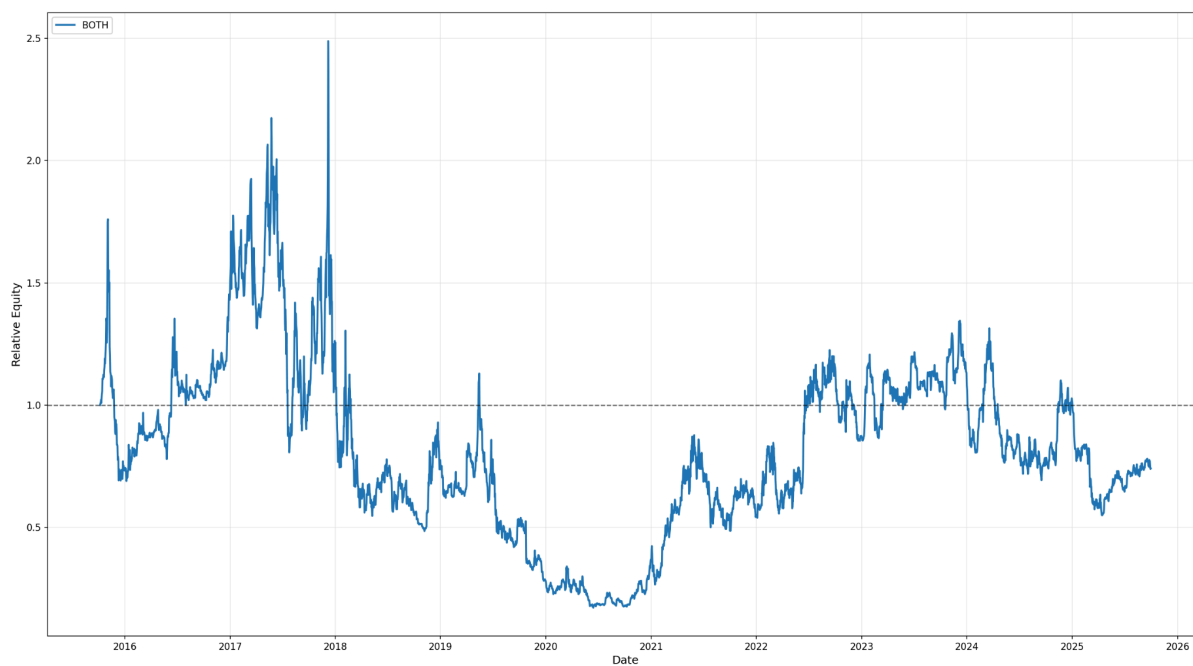
SELLONLY (1D)

Annualized Return: **-33.79 %**
Maximum Drawdown: **-98.95 %**
Annualized Volatility: **48.35 %**
Sharpe Ratio: **-0.70**
Calmar Ratio: **-0.34**
Number of Trades: **292**



BOTH (1D)

Annualized Return: **-2.70 %**
Maximum Drawdown: **-93.14 %**
Annualized Volatility: **67.46 %**
Sharpe Ratio: **-0.04**
Calmar Ratio: **-0.03**
Number of Trades: **584**



5.3 Complete Performance Metrics (All Strategies)

Bitcoin Buy & Hold (Benchmark)

Timeframe: 1H

Annualized Return: 84.99 %

Max Drawdown: -84.20 %

Volatility: 72.86 %

Sharpe: 1.17

Calmar: 1.01

Trades: 0

Timeframe: 1D

Annualized Return: 75.28 %

Max Drawdown: -83.30 %

Volatility: 67.34 %

Sharpe: 1.12

Calmar: 0.90

Trades: 0

PSAR BUYONLY

Timeframe: 1H

Annualized Return: 35.95 %

Max Drawdown: -81.83 %

Volatility: 49.69 %

Sharpe: 0.72

Calmar: 0.44

Trades: 6,370

Timeframe: 1D

Annualized Return: 46.94 %

Max Drawdown: -66.73 %

Volatility: 46.96 %

Sharpe: 1.00

Calmar: 0.70

Trades: 292

PSAR SELLONLY

Timeframe: 1H

Annualized Return: -44.55 %

Max Drawdown: -99.74 %

Volatility: 53.27 %

Sharpe: -0.84

Calmar: -0.45

Trades: 6,382

Timeframe: 1D

Annualized Return: -33.79 %

Max Drawdown: -98.95 %

Volatility: 48.35 %

Sharpe: -0.70

Calmar: -0.34

Trades: 292

PSAR BOTH

Timeframe: 1H

Annualized Return: -24.62 %

Max Drawdown: -97.75 %

Volatility: 72.85 %

Sharpe: -0.34

Calmar: -0.25

Trades: 12,752

Timeframe: 1D

Annualized Return: -2.70 %

Max Drawdown: -93.14 %

Volatility: 67.46 %

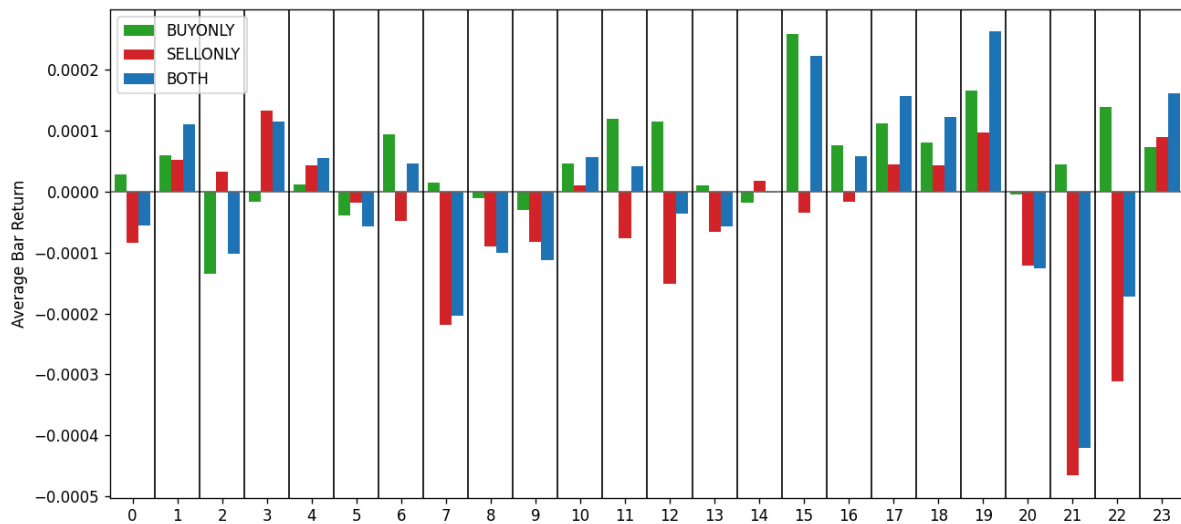
Sharpe: -0.04

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Trades: 584

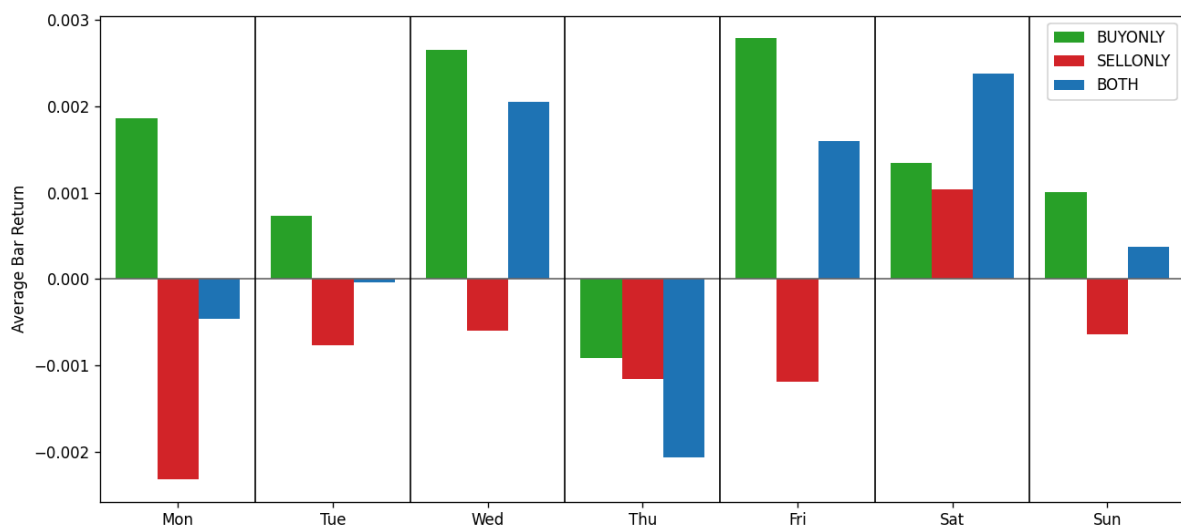
6. Seasonality Analysis

6.1 Hour-of-Day Seasonality (1H)



BUYONLY displays the strongest positive returns between **15:00–19:00**, peaking at **18:00–19:00**.

6.2 Day-of-Week Seasonality (1D)



Best days: **Monday, Wednesday, Friday**

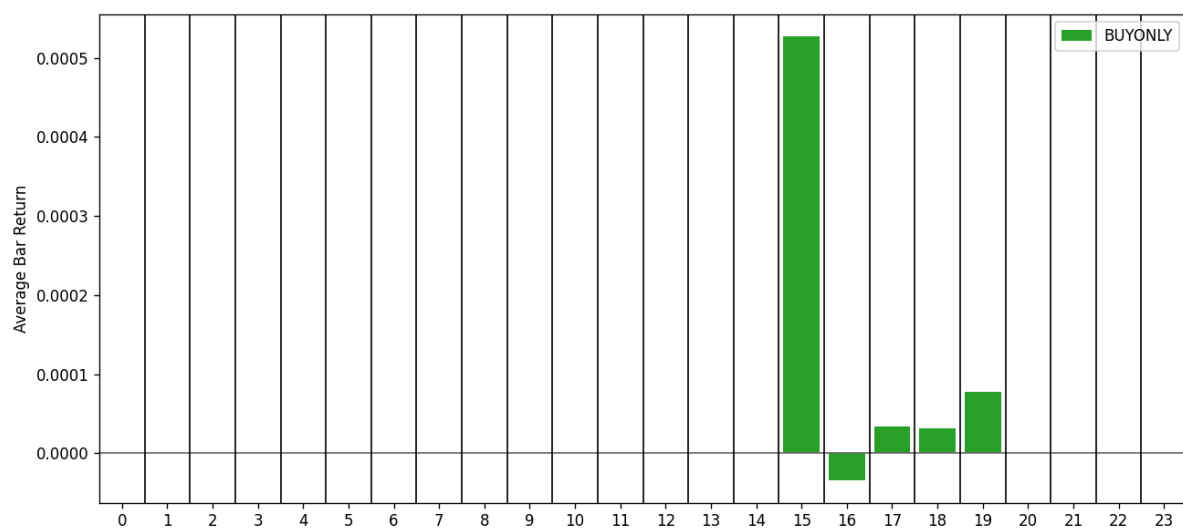
Worst days: **Thursday, Sunday**

7. Calibrated SAR Model (BUYONLY)

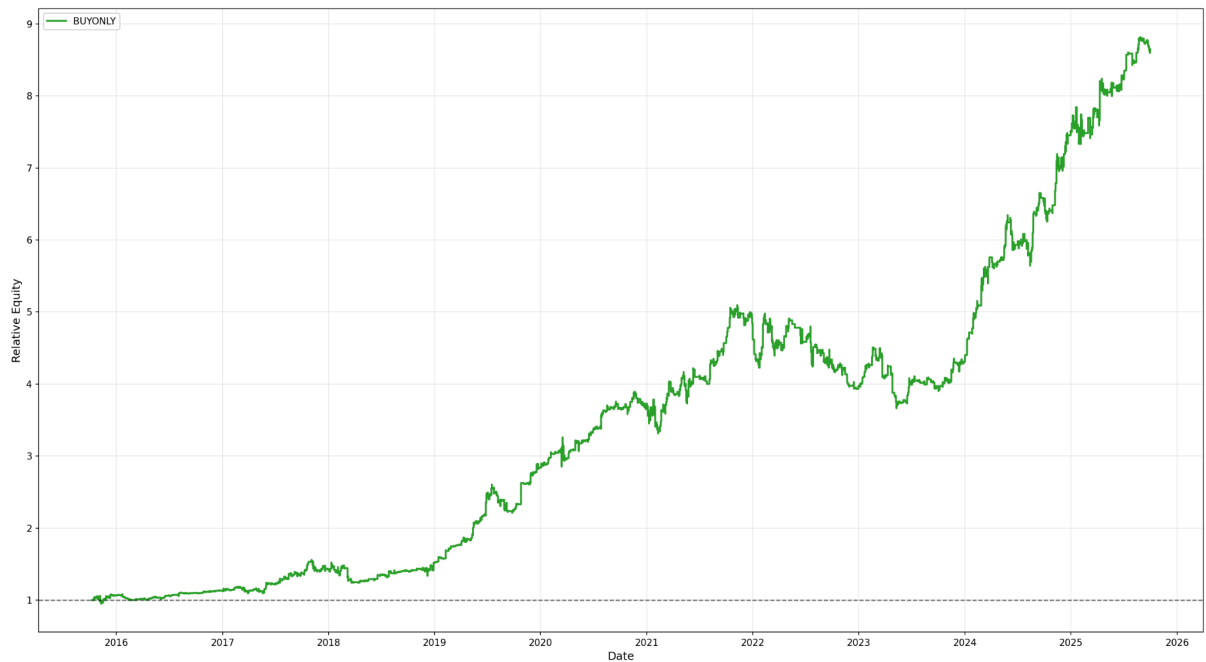
Calibration Filters Applied

- Trade only on: **Monday, Wednesday, Friday**
- Trade only between **15:00–19:00 UTC**
- Long-only

BUYONLY Hourly Profitability Basis



7.3 Calibrated BUYONLY Equity Curve



7.4 Performance Metrics (Calibrated)

Timeframe: 1H

Annualized Return (%): 24.11

Max Drawdown (%): -28.11

Annualized Volatility (%): 17.83

Sharpe Ratio: 1.35

Calmar Ratio: 0.86

Number of Trades: 1978

7.5 Interpretation

The calibrated BUYONLY Parabolic SAR model demonstrates a substantial improvement in stability, efficiency, and risk-adjusted performance compared to the raw versions of PSAR.

Restricting trading activity to historically profitable time windows (Monday, Wednesday, Friday between 15:00–19:00 UTC) fundamentally alters the behaviour of the strategy.

First, the risk-adjusted performance increases dramatically. The Sharpe ratio rises from **0.72** in the unrestricted 1H BUYONLY model to **1.35** in the calibrated version, marking the highest Sharpe ratio among all PSAR configurations examined in this study. A Sharpe above 1.0 reflects a robust and consistent return profile, and a value above 1.3 suggests institutionally attractive performance for a simple, long-only trend-following system.

Second, the calibrated approach substantially reduces risk. Maximum drawdown declines from **−81.83 %** to **−28.11 %**, a reduction of more than **50 percentage points**, while annualized volatility falls from **49.69 %** to only **17.83 %**. These improvements stem from avoiding low-liquidity trading hours, bypassing noise-heavy regimes, and eliminating SAR-induced whipsaws outside of well-performing intraday windows.

Third, despite reduced market exposure—approximately **12 hours per week**—the annualized return remains positive (**24.11 %**). This demonstrates that Parabolic SAR signals possess exploitable structure when constrained to periods where BTC historically exhibits directional momentum.

Overall, the calibrated model transforms SAR from a reactive, unstable indicator into a **risk-efficient and selectively profitable trend-following strategy**. While the return is lower than long-term Buy & Hold, the superior risk-adjusted metrics (Sharpe 1.35, Calmar 0.86) validate the effectiveness of combining PSAR signals with structural timing filters. Nevertheless, the strategy still lacks regime-awareness and would likely benefit from complementary filters such as ATR volatility expansion, EMA-based trend confirmation, or higher-timeframe directional bias.

8. Discussion

Main insights:

1. SAR is overly reactive → extreme whipsaws.
2. SELLONLY is structurally non-viable on BTC.
3. BOTH amplifies noise → catastrophic drawdowns.
4. Daily timeframe is more stable but still inferior to Buy & Hold.
5. Time-window calibration improves stability but not profitability.
6. SAR requires additional filters to become viable:
 - ATR volatility filter
 - EMA200 higher-timeframe trend
 - D1→H1 regime model
 - or multi-indicator confirmation

9. Conclusion

This study demonstrates that the Parabolic SAR indicator, when applied to Bitcoin from 2015 to 2025, performs poorly as a continuous standalone system. The unrestricted BUYONLY, SELLONLY, and BOTH modes all suffer from deep drawdowns, frequent whipsaws, and unstable equity curves. These findings confirm that PSAR, by itself, is overly reactive in a market characterized by sharp volatility cycles and structural bullish drift.

However, the seasonality-calibrated BUYONLY model reveals that SAR can become **meaningfully effective** when combined with structural timing constraints. Restricting trades to historically profitable days and hours transforms the behavior of the strategy: volatility decreases sharply, drawdowns shrink, and the Sharpe ratio increases to **1.35**, with a Calmar ratio of **0.86**. These metrics are competitive with institutional-quality trend-following systems and indicate that Parabolic SAR contains a usable trend signal when noise is filtered out.

The key insight is that **SAR alone is not the problem — trading it continuously is**. Bitcoin's intraday volatility structure rewards selective exposure, and time-based filters help isolate the conditions under which PSAR signals have predictive value. Even so, full system viability will require additional layers: volatility regime detection, higher-timeframe trend confirmation, or complementary indicators that reduce SAR's sensitivity to noise.

Future research should integrate PSAR into multi-factor frameworks, including ATR filters, EMA200 trend bias, D1→H1 confirmation, or price-structure models. These enhancements may elevate the calibrated strategy into a fully robust, deployable trading system.

10. References

Wilder (1978). *New Concepts in Technical Trading Systems*.

Appendix (Figures)

All computational results, backtests, performance metrics, equity curves, and seasonality analyses presented in this study were independently generated by the author.

All data processing, modeling, visualization, and research procedures were executed directly by the author using custom Python research scripts.

No external analytical services or third-party model outputs were used.